Reader's Guide to the Relational Carrier Theory Corpus

(P0 — canonical synthesis of Papers 1–6 and the bridge note)

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Abstract

This document is the canonical entry point to the relational-carrier-theory corpus. It is a synthesis, not a paper: every claim cites the load-bearing source paper, no new numerical result or proof is introduced here. The corpus consists of six sector papers (P1–P6) plus a bridge note; this guide states the theory's one-sentence thesis, identifies the three load-bearing claims, maps each paper to its role, registers the closure grammar (EXACT / PRECISE / FACTOR2; load-bearing / downstream / supporting / external / open), tabulates the top-tier closures, and lists the falsification handles that would settle the framework's main predictions within the next ~ 2 years. The intended reader is either a physicist arriving at the corpus for the first time or a reviewer wanting the architecture before the technical layers.

What this document is and is not

What this is. A reader's guide. A map of where each question is settled in the corpus. A registry of the closure grammar. A pre-organised table of the load-bearing claims and falsification handles.

What this is not. Not a paper. Not a derivation. Not a declaration of completeness. Not peer-reviewed. The relational carrier theory is a work-in-progress closure construction by an independent researcher; the empirical and conceptual claims of Papers 1–6 are presented as candidate closures whose external validation is the principal next step. *No claim made here is novel*: every numerical value, every closure form, every falsification handle is sourced to the technical paper that owns it. If a sentence in this guide and a sentence in the cited paper disagree, the cited paper is authoritative.

1 One-sentence thesis and short genealogy

One-sentence thesis. Physical reality is described as a finite relational carrier whose correlation structure generates the four observable channels of the bounded-operator readout — two primary (Ξ-cohesion, Ω-synchronisation) and two coupled (Ξ-ψ common, sync-floor) — from which the System-$\mathcal{R}$ rational coefficient tuple

$$(\gamma, \alpha_\xi, \varepsilon^2_{\text{sync}}, \beta_\pi, D_\Omega) \;=\; (1/10,\ 9/10,\ 1/20,\ 15/16,\ 67/80)$$

is read off, and from which the emergent geometry of general relativity, the Standard-Model fermion-mass-and-mixing spectrum, dark-sector phenomenology, and the Bekenstein–Hawking entropy $S/A = 1/4$ follow as algebraic consequences without additional free parameters.

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Foundational note: relation precedes information. The corpus's architecture instantiates a logical priority that is worth stating explicitly. A difference $a \neq b$ is not information in isolation: it requires a comparison relation in which a and b are mutually accessible. The relational carrier supplies that comparison structure as its primitive datum; the System-$\mathcal{R}$ rational tuple, the bounded-operator channels, and all downstream observables are read off from it *after* the relational structure is in place, not before. In this construction the name "relational carrier theory" is literal: *the relations come first, the differences become information by virtue of the relations*, and emergent geometry and physical observables are algebraic consequences of the relational correlation structure. The empirical Lemma B finding that the spectral gap λ_2 is globally distributed — not localised on any defect sub-population (P4 [4]) — is the operational counterpart of this priority: a connected relational carrier is the precondition for any observable to be *information about* anything else.

One-paragraph genealogy. The System-$\mathcal{R}$ primitive set is itself fully determined by two integer inputs — the relational-spacetime dimension $d = 4$ and the fermion generation count $N_{\text{gen}} = 3$ — via $\gamma = 1/(2(d{+}1))$, $\alpha_\xi = (2d{+}1)/(2(d{+}1))$, $\varepsilon^2_{\text{sync}} = 1/(4(d{+}1))$, $\beta_\pi = 15/16$, $D_\Omega = 1-(N_{\text{gen}}d{+}1)/(d^2(d{+}1))$. Both d and N_{gen} are observationally fixed integer counts of discrete-geometric features of the underlying relational lattice, not free parameters. The free-parameter count of the structural input is therefore zero. The five System-$\mathcal{R}$ rationals are the algebraic backbone of all downstream closures.

Derived primitive identities. Several closed-form identities among the primitives recur across the corpus and are used repeatedly in the downstream closures. For convenience of cross-paper reading they are registered here:

- $\alpha_\xi + \gamma = 1$ (chirality complementarity; P2 § 5);
- $\varepsilon^2_{\text{sync}} = \gamma/2$ (fluctuation–dissipation symmetry; P2 § 5);
- $s_{\text{face}} = 1/d = 1/4$ (Bekenstein–Hawking entropy face, equivalently the inverse-dimension prefactor; P3 L7);
- $d + N_{\text{gen}} = 7$ (mass-mode dimensional primitive; recurs in $L4 = -(d{+}N_{\text{gen}})^2/(d(d{+}1)) = -49/20$, $n_s = 1 - \gamma^2(d{+}N_{\text{gen}})/2 = 193/200$, and $Y_p = (d{+}N_{\text{gen}})^2\gamma^2/2 = 49/200$);
- $2(d{+}1) = 10 = \gamma^{-1}$ (back-channel inverse; the leading integer in the Coleman–Gamow action $S_{\text{Gamow}} = 2(d{+}1)N_{\text{gen}} - (2d{+}N_{\text{gen}}) = 19$);
- $2d + N_{\text{gen}} = 11$ (CKM dimensional primitive; $V_{cb} = \alpha_\xi/(2(2d{+}N_{\text{gen}})) = 9/220$);
- $(d{+}1)/d = 5/4$ (gravitational-backreaction screening amplitude; cosmological-constant Schwinger–Dyson layer);
- $N_{\text{gen}} \cdot d = 12$ (occupancy-sector dimensional product; pairs-per-cell on the canonical ladder).

Each identity is a closed-form consequence of the integer inputs (d, N_{gen}); they are not independent free parameters.

2 Three load-bearing claims

The technical-foundations paper P6 [9] states the load-bearing classification explicitly in its "three claims are load-bearing" paragraph: the construction stands on

1. the *bounded carrier with finite-N Einstein-residual closure* of P4 [4] (emergent gravity, lattice $\Lambda_{\rm lat} = 19/15$ identity, four-point Continuum Backreaction Identity);
2. the *System-$\mathcal{R}$ coefficient readout via the bounded-operator construction* of P2 [2] (ten causal-wave landings, formula-search audit, the four-projector spectral uniqueness theorem);
3. the *topology-labeled loop-class library mapping* of P3 [3] (loop-class protocol, Yukawa-cluster PROSP-01/04, Bekenstein–Hawking $S/A = 1/4$).

All other cross-sector closures — $\alpha_{\rm EM}$, $v_{\rm EW}$, $S_{\rm BH}/A = 1/4$, n_s, A_s, halo phenomenology, dark sector — are *downstream* of the three load-bearing claims in the sense defined in Section 5.

3 Two branches: vacuum and matter-side chirality flip

The bridge note [10] formulates the corpus-wide two-branch structure as a single *loop-topological carrier* operating in two distinct relational regimes:

Vacuum branch. Pre-chirality-flip configuration ($\theta_{\rm chir} = 0$, canonical $(\alpha_\xi, \gamma) = (9/10, 1/10)$). Governs the electroweak-scale closure ($v_{\rm EW} = 246.22$ GeV via P1 [1]'s four-path consensus), the inflation-era and recombination-era observables (n_s, A_s, z_*, θ_*), and the late-time dark-energy equation of state at leading order ($w_{\rm DE} = -39/40$).

Matter-side chirality-flip branch. Post-$\pi/4$ matter-side branch with asymptotic/inverted anchor $(\alpha_\xi, \gamma) = (1/10, 9/10)$: the chirality-flip boundary itself sits at $\theta_{\rm chir} = \pi/4$ (where the two coefficients cross), and the asymptotic post-flip anchor is the inverted tuple $(1/10, 9/10)$ approached as $\theta_{\rm chir}$ continues past $\pi/4$. This branch governs the matter-core phenomenology (top-5% K/Q closure), the chirality-running $w_{\rm DE}(\theta_{\rm chir})$ at next-to-leading order (P4B [6]), and the absolute neutrino-mass identification $m_1 = \gamma^2 m_3$ via the loop-class companion's Pure-Self-Energy resummation (P3 [3]).

The two-branch decomposition is itself System-$\mathcal{R}$-derived rather than ad hoc: the chirality angle $\theta_{\rm chir}(N)$ runs algebraically with lattice resolution N [4] (the running formula is the one labelled *theta-running-first-principles* in P4), crossing $\pi/4$ at a specific lattice resolution at which the back-channel-projection numerical hierarchy inverts.

4 Corpus map: which paper does what

5 Claim grammar

The corpus uses a registered vocabulary of closure tiers and claim classes. Each is defined formally in one of the technical papers; the present section is a routing table only.

Closure tiers (precision against external anchor).

- **EXACT**: relative residual at or below the paper-specific EXACT cut, corresponding to a System-$\mathcal{R}$ identity that closes at machine precision under the rational reduction $\mathcal{R}$ (see P6 §1 definition).
- **PRECISE**: relative residual $\leq 2.5\%$; closes the framework prediction within standard-fit experimental uncertainty.
- **FACTOR2**: closure form correct but numerical value within a factor of 2; reported as a structural-form identification, not a precision claim.

Paper	Title (short)	Role in the corpus
P0 (this doc.)	Reader's guide	Entry-point, claim grammar, corpus map, falsifiers
P1 [1]	Hierarchy-bridge electroweak-scale closure	$v_{\mathrm{EW}} = 246.22$ GeV four-path consensus, $\lvert g\rvert_{\mathrm{MS}} = 1.42$, HBR + $\theta_{em} = \pi$ load-bearing input to P4
P2 [2]	Causal-wave landings + System-$\mathcal{R}$ readout	**Pillar (ii)**: bounded-operator construction; ten landings; four-projector spectral-uniqueness theorem; formula-search audit $47/{\sim}174$k
P3 [3]	Loop-class library + closure protocol	**Pillar (iii)**: loop-class topology-labelled mapping; Bekenstein–Hawking $S/A = 1/4$; SYE Yukawa/CKM/PMNS pipeline; PROSP-01/04 prospective registry
P4 [4]	Emergent-gravity bulk Einstein closure	**Pillar (i)**: bounded carrier; $\Lambda^{\infty}_{\mathrm{lat}} = 19/15$; Continuum Backreaction Identity; closure-domain ladder $N \in [50, 512]$
P4A [5]	Schwarzschild + PPN companion	Companion test: weak-field PPN parameters; $\bar{\theta}_{\mathrm{QCD}}$ vanishing
P4B [6]	Anisotropic source → DM/DE companion	Companion: $T^{\Xi}_{\mu\nu}$ decomposition; $w_{\mathrm{DE}} = -39/40$; chirality-running $w(z)$; 9-layer cosmological-constant dressing
P4C [7]	H3c witnesses companion	Companion diagnostics for the Einstein-residual closure
P4D [8]	Atiyah–Singer chirality index companion	Companion: integer-quantised vortex-DM winding number; topological-DM identification
Bridge note (appendix) [10]	P1↔P4 cross-file consistency	Lemma-level: common loop-topological carrier; version-lock matrix; cross-paper $\mathcal{R}$-tuple consistency
P6 [9]	Relational-UV-closure technical foundations	Master technical document: H_{UV}, S_{UV}, Z_{UV}, branch construction, closure ledger, load-bearing-vs-downstream registry

Table 1: Corpus map. P0 (this document) is the reader's guide and entry point; P2, P3, P4 are the three load-bearing claims of Section 2; P1, P4A–D, and P6 supply supporting structure or technical foundations; the bridge note is a cross-file consistency appendix.

- **STRUCTURAL_CLOSED**: closure form derived algebraically; numerical match secondary (e.g. $\Sigma_f Y_f^2 = 10$).

Paper-specific EXACT cuts. The corpus uses three EXACT cuts depending on the observable class. The present routing table makes them explicit so a reader cross-comparing two papers does not confuse a stricter EXACT-label in one paper with a looser one in another. A claim that is EXACT under a stricter cut is automatically EXACT under any looser cut (the implication runs from left to right).

Paper / observable class	EXACT cut	Rationale
P2 (causal-wave landings)	$\lvert r\rvert \le 0.01\%$	EXACT-target algebraic landings; sub-experimental-uncertainty closure
P3 (loop-class library)	$\lvert r\rvert < 0.4\%$	loop-corrected observables compared against PDG/NuFIT measurement bands
P0 / P4 / P4A–D / P6 / bridge	$\lvert r\rvert \le 1\%$	global manifest band covering all observable classes; superset of P2 and P3 cuts

The PRECISE cut is uniform across the corpus ($\lvert r\rvert \le 2.5\%$) and the FACTOR2 cut is uniform ($\lvert r\rvert \le 100\%$, i.e. within a factor of two). A claim that closes at $\lvert r\rvert \le 0.01\%$ is EXACT under any of the three cuts; a claim at $\lvert r\rvert \in (0.01\%, 0.4\%]$ is EXACT under P3, P0 and the global manifest but downgrades to PRECISE under P2; a claim at $\lvert r\rvert \in (0.4\%, 1\%]$ is EXACT under P0 and the global manifest but PRECISE under P3.

Claim classes (load-bearing classification). The unified manifest (file `data/corpus_claims_manifest.json` in P6's repository [9]) registers every main claim under one of five labels:

- **Load-bearing.** The construction stands on this claim. Three exist: pillars (i)–(iii) of Section 2.
- **Downstream.** An algebraic or audit consequence of the same System-$\mathcal{R}$ tuple read off by the load-bearing claims (e.g. $\alpha_{\mathrm{EM}} = \gamma^2 \alpha_\xi^{N_{\mathrm{gen}}}$, $\sigma_8 = \alpha_\xi^2$, $n_s = 1 - 7\gamma^2/2$). The vast majority of the closures of Table 2 fall here.
- **Supporting witness.** An independent audit diagnostic that does not extend the closure surface but confirms it (e.g. P4C's H3c gravitational-coupling-scaling diagnostics).
- **External contact.** A point of contact with the existing literature (Pauli–Zeldovich cancellation, Schwinger–Dyson resummation, Reed–Simon VIII.5).
- **Open target.** A registered observable whose closure is not yet established within the corpus and which is flagged as the framework's primary near-term theoretical target.

6 Main closures at a glance

The list below is a curated subset of the ~ 50 registered closures of the corpus, selected for highest external-anchor visibility. Each entry is owned by the cited paper; this table re-states them in one place.

Table 2: Top-tier corpus closures, one row per observable. "Tier" is the precision class against the external anchor (EXACT $\leq$ 1%, PRECISE $\leq$ 2.5%, FACTOR2 within 2×). "Class" is the load-bearing classification of Section 5. Rows marked *load-bearing* are components of the three load-bearing blocks (pillars (i)–(iii) of Section 2), not additional independent pillars; the corpus has *exactly three* load-bearing pillars and the multiplicity of load-bearing rows in the table reflects the multiple observables that each pillar supplies.

Observable	System-$\mathcal{R}$ form	Tier	Class	Owner
	Cosmology			
H_0	$100\,\alpha_\xi\, s_{\text{face}}\, N_{\text{gen}} = 67.5$ km/s/Mpc	EXACT	downstream	P4B
σ_8	$\alpha_\xi^2 = 81/100$	PRECISE	downstream	P4B
S_8	$\alpha_\xi^2\sqrt{\Omega_m/0.3} = 0.828$	EXACT	downstream	P4B
Ω_m	$47/150$	PRECISE	downstream	P4B
Ω_Λ	$103/150$	PRECISE	downstream	P4B
$\Omega_{\text{DM}}h^2$	$243/2000 = 0.1215$	PRECISE	downstream	P4B
w_{DE}	$-1 + \varepsilon_{\text{sync}}^4/\gamma = -39/40$	PRECISE	downstream	P4B
n_s	$1 - 7\gamma^2/2$	EXACT	downstream	P4B
A_s	$21\gamma^{10}$	EXACT	downstream	P4B
z_*	$(2d+1)(2d+N_{\text{gen}})^2$+sub	EXACT	downstream	P4B
Y_p	$49\gamma^2/2$	EXACT	downstream	P4B
Σm_ν	$\gamma^2 m_3 + m_2 + m_3 \simeq 0.060$ eV	PRECISE	downstream	P6
	Electroweak & Standard Model			
v_{EW}	4-path consensus 246.22 GeV	PRECISE	load-bearing input	P1
α_{EM}	$\gamma^2\alpha_\xi^{N_{\text{gen}}} = 729/100{,}000$	EXACT	downstream	P3
$\sin^2\theta_W$	$1/d - \gamma/(2N_{\text{gen}}) = 7/30$	PRECISE	downstream	P3
V_{us}	$\alpha_\xi\, s_{\text{face}} = 9/40$	EXACT	downstream	P3
m_t	spectral Yukawa eigenvalue $\to$ 174.1 GeV	EXACT	downstream	P3
m_μ/m_e	spectral Yukawa chirality split $\to$ 206.0	EXACT	downstream	P3
$a_\mu^{\text{Schwinger}}$	$\alpha_{\text{EM}}/(2\pi)$	PRECISE	downstream	P3
	Gravity & black-hole sector			
$\Lambda_{\text{lat}}^\infty$	$3\alpha_\xi/2 + \varepsilon_{\text{sync}}^2 - \gamma(N_{\text{gen}}+1)/N_{\text{gen}} = 19/15$	EXACT	load-bearing	P4
Λ_t^∞	$\alpha_\xi^2 = 81/100$	EXACT	load-bearing	P4
Λ_s^∞	$-\gamma^2/2 = -1/200$	EXACT	load-bearing	P4
S_{BH}/A	$s_{\text{face}} = 1/d = 1/4$	EXACT	load-bearing	P3

Observable	System-$\mathcal{R}$ form	Tier	Class	Owner
PPN γ_{PPN}	induced from Einstein closure	PRECISE	downstream	P4A
$\rho_\Lambda^{\mathrm{pred}}/\rho_\Lambda^{\mathrm{obs}}$	9-layer dressing $\to 1.001$	EXACT	downstream	P4B
		Dark sector & halo phenomenology		
$g_\dagger$ (RAR)	$cH_0/(2\pi\alpha_\xi)$	PRECISE	downstream	P4B
BTFR A	$\sim(G_N H_0)^{-1}$	PRECISE	downstream	P4B
NFW vs Burkert branching	polarity-violation rate	STRUCTURAL	downstream	P4B
SPARC pass-fraction	79/129 within domain $\mathcal{D}_{\mathrm{halo}}$	PRECISE	downstream	P4B
Vortex-DM w_{DM}	0 (integer-quantised winding)	STRUCTURAL	downstream	P4D
		Foundations		
4-projector basis	Reed–Simon VIII.5 spectral-uniqueness	EXACT	load-bearing	P2
10 causal-wave landings	47 forms / $\sim$ 174k searched	EXACT	load-bearing	P2
Loop-class library	topology-labelled protocol	STRUCTURAL	load-bearing	P3
$\sum_f Y_f^2 = 10$	three-generation chiral content	EXACT	downstream	P6
$\bar{\theta}_{\mathrm{QCD}} = 0$	APS-chirality pairing	STRUCTURAL	downstream	P4A

7 Falsification handles

The framework registers explicit falsification triggers against current and near-term cosmological and laboratory data. Closure of these handles within the next $\sim$2 years is the principal external test of the framework.

Cross-reference to owning-paper falsifier IDs. The six corpus-wide falsifiers below map to the owning-paper local falsifier labels as registered in each companion paper's `Falsification triggers` section.

P0 falsifier (this section)	Owning paper	Local label
1. Σm_ν vs DESI / CMB-S4 / Roman / Euclid	P3, P6	P3 FAL-Σm_ν, P6 §FAL-cosmo
2. Dynamical $w(z)$ vs DESI BAO running	P4B	P4B FAL-w_a
3. Hubble tension on non-Cepheid distance ladders	P4B	P4B FAL-H_0
4. Tensor-to-scalar ratio r vs CMB-S4	P4B	P4B FAL-r
5. Particle-DM direct detection (LZ / XENON-nT / PandaX / DarkSide)	P4D	P4D F1, F2, F3
6. Halo regime signatures (microlensing, EDGES 21-cm)	P4, P4B	P4B FAL-halo

A reader following one of these falsifiers across to its owning paper finds the same numerical thresholds at higher technical detail and the reproducer pointers.

Cosmological-data-decidable within ~ 2 years.

1. *Σm_ν vs DESI / CMB-S4 / Roman / Euclid.* The framework predicts $\Sigma m_\nu \simeq 0.060$ eV via $m_1 = \gamma^2 m_3$ [3, 9]. The current DESI DR2 BAO + DR1 full-shape bound is $\Sigma m_\nu < 0.0642$ eV [11]; the framework prediction sits 6.1% below this. A confirmed tightening below the NH oscillation minimum $\simeq 0.0586$ eV would falsify the $\gamma^2 m_3$ identification.

2. *Dynamical $w(z)$ vs DESI BAO running.* The framework predicts $|w_a| \lesssim \gamma^2 = 0.01$ from the chirality-running structure $w_{\rm DE}(\theta_{\rm chir})$ [6]. The DESI 2024+2025 central $w_a \approx -0.4$ at $\sim 3\sigma$ lies $\sim 40\times$ outside the framework band; consolidation to $> 5\sigma$ in DESI-DR3 would falsify the $\varepsilon^4_{\rm sync}/\gamma$ closure.

3. *Hubble tension on non-Cepheid distance ladders.* The framework predicts $H_0 = 67.5$ km/s/Mpc independent of distance-ladder method [6]. Confirmed $H_0 \in [72, 74]$ km/s/Mpc on a non-Cepheid local ladder (gravitational-wave sirens, TDCOSMO, JWST-anchored TRGB at $\sigma_{H_0} \lesssim 1$ km/s/Mpc) would falsify the distance-ladder-systematic attribution of the SH0ES 73.04 ± 1.04 discrepancy.

4. *Tensor-to-scalar ratio r vs CMB-S4.* The framework predicts $r \sim \gamma^3(d + N_{\rm gen}) \simeq 7 \times 10^{-3}$ within the BICEP/Keck BK18 upper bound $r_{0.05} < 0.036$ [12]. CMB-S4 sensitivity $\sigma(r) \sim 10^{-3}$ within ~ 3 years will provide a definitive detection or upper-bound test.

Laboratory-data-decidable.

5. *Particle-DM direct detection.* The framework identifies dark matter with a relational vortex-defect network of integer-quantised $U(1)$ winding [8]; this is topological, not particulate. The framework therefore predicts *null* results on LZ ($\sigma_{\rm SI} < 2.2 \times 10^{-48}$ cm^2 at 40 GeV/c^2 [13]), XENON-nT [14], PandaX-4T [15], and DarkSide [16] down to the neutrino floor. A positive nuclear-recoil signal at any future facility would falsify the topological-DM identification.

6. *Halo regime signatures.* Microlensing of vortex-network clumps at the chirality-flip mass scale $M_{\rm flip}$ [4] should show a discrete event-frequency distribution rather than a continuous PBH spectrum; 21-cm absorption-depth deviations from ΛCDM in the EDGES band correlate with the vortex-density power spectrum at $z \sim 17$.

Pre-registered prospective targets. The loop-class companion paper P3 [3] registers a prospective cluster registry — the file `prospective_cluster_registry.json` in the loop-class repo — that pre-registers three loop-class observables for future $\mathcal{R}$-falsification opportunities at the P3 protocol level.

8 Where to read next

The natural reading order depends on the reader's question:

"What is the theory?". Read this guide (P0) first. Then read P6 [9] for the technical foundations, P2 [2] for the System-$\mathcal{R}$ readout, P3 [3] for the loop-class protocol, and P4 [4] for the main emergent-GR claim. The bridge note [10] is best read afterward as a cross-file consistency appendix.

"Where do the System-$\mathcal{R}$ coefficients come from?". P2 [2] (causal-wave landings + formula-search audit + spectral-uniqueness theorem).

"How does general relativity emerge?". P4 [4] (bounded carrier + Einstein-residual closure + Continuum Backreaction Identity); then P4A [5] for the weak-field PPN companion.

"How do the Standard-Model masses and mixings close?". P3 [3] (loop-class library + SYE Yukawa-pipeline + α_{EM}-mediated identities).

"How do dark matter and dark energy fit in?". P4B [6] (anisotropic source decomposition, w_{DE}, cosmological-constant dressing) and P4D [8] (vortex-DM topological identification).

"What is the electroweak-scale closure?". P1 [1] (HBR + $\theta_{em} = \pi$ four-path consensus for v_{EW}).

"Is the corpus internally consistent?". The bridge note [10] (technical-consistency proof P1↔P4) and the unified manifest file `relational-uv-closure-repro/data/corpus_claims_manifest.json` of P6 [9].

Closing note

The relational carrier theory is presented as a candidate closure construction whose external validation is the principal next step. The corpus does not claim to be a final accepted theory of nature; it claims that the load-bearing pillars (i)–(iii) of Section 2 are internally consistent, that the downstream closures of Table 2 follow algebraically from the load-bearing pillars within the System-$\mathcal{R}$ rational reduction, and that the falsification handles of Section 7 provide a decidable near-term test surface. Each individual claim is owned by the cited paper; the present guide is an entry point and a map, nothing more.

Data and software availability

The reproducibility package (https://github.com/TerraSignum/relational-carrier-manifest) bundles all input data files (under `data/`), reproducer scripts (`src/verify_*.py` and `src/make_figures.py`), and unit tests (`tests/`). Cryptographic provenance is provided by `data/SHA256SUMS` (GNU `sha256sum -c` compatible), and the publication-prep frozen state is documented in `RELEASE_v1_0_0.md` with full release notes in `CHANGELOG.md`. Software metadata is in `pyproject.toml` and `CITATION.cff` (Citation File Format 1.2). The package is licensed under MIT; the manuscript text is licensed under CC-BY-4.0.

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